

Wheels And Tires



Safety Precautions

WARNING: Vehicle may have multiple drive wheels. Do not use engine to power the driveline unless all drive wheels are elevated off the ground. Drive wheels in contact with ground could cause unexpected vehicle movement. Failure to follow this instruction may result in serious personal injury.

WARNING: Always match the tire size to the wheel size during assembly. Incorrect matching can result in tire bead damage or tire separation from the wheel. Failure to follow this instruction may result in serious personal injury to technician or vehicle occupant(s).

WARNING: Before servicing any tire, ask the customer if anyone injected a tire sealant into the tire. Tire sealants may be flammable and can burn or explode if exposed to an ignition source. Failure to follow this instruction may result in serious personal injury.

WARNING: Replacement wheels must be equivalent to the original equipment wheels in:

- load carrying capacity.
- diameter, width and offset.
- pilot hole and bolt circle.

Combined load carrying capacity of replacement wheels for a given axle, must be equal to or greater than that axle's gross axle weight rating (GAWR) identified on the vehicle's Safety Compliance Certification label. All other specifications should be evaluated by measurement of both the original wheel and the replacement wheel. If specifications are not equivalent, the safety and handling of the vehicle may be degraded, which may result in serious injury to the vehicle occupant(s).

WARNING: Wear eye and ear protection when servicing a vehicle. Failure to follow this instruction may result in serious personal injury.

WARNING: Keep eyes away from valve stem when deflating tires. Reduce air pressure in tire as much as possible by pushing in valve core plunger prior to removing the core. Escaping air can carry particles that can injure the eyes. Failure to follow these instructions may result in serious personal injury.

WARNING: Only use replacement tires that are the same size, load index, speed rating and type (such as P-metric versus LT-metric or all-season versus all-terrain) as those originally provided by Ford. The recommended tire and wheel size may be found on either the Safety Compliance Certification Label or the Tire Label, which is located on the B-pillar or edge of the driver's door. If the information is not found on these labels, consult a Ford dealer. Use of any tire or wheel not recommended by Ford can affect the safety and performance of the vehicle, which could result in an increased risk of loss of vehicle control, vehicle rollover, personal injury and death. Additionally, the use of non-recommended tires and wheels could cause steering, suspension, axle or transfer case/power transfer unit failure.

NOTICE: Do not clean aluminum wheels with steel wool, abrasive-type cleaners or strong detergents. Damage to the wheel finish may occur. Use Wheel and Tire Cleaner ZC-37-A or -B or equivalent.

Factory-installed tires and wheels are designed to operate satisfactorily with loads up to and including full-rated load capacity when inflated to recommended inflation pressures.

Correct tire pressure and driving techniques have an important influence on tire life. Heavy cornering, excessively rapid acceleration and unnecessary sharp braking increase tire wear.

To equalize tire wear, the tires should be rotated at recommended intervals.

Tire Pressure Monitoring System (TPMS)

The Tire Pressure Monitoring System (TPMS) includes:

- the Body Control Module (BCM) , located behind the passenger side kick panel.
- the Tire Pressure Monitor (TPM) module, located behind the instrument panel on the RH side near the glove box.
- four valve stem mounted tire pressure sensors.
- an Instrument Panel Cluster (IPC) warning indicator.
- message center warnings.

Tire Pressure Monitor (TPM) Module

The TPM module is a radio receiver that collects the tire pressure data from the TPMS tire pressure sensors. The TPM module passes the tire pressure data to the BCM , where the BCM applies a predetermined pass/fail criteria.

The BCM compares the information of each tire pressure message against a pressure limit. If the BCM determines that the tire pressure has fallen below the minimum pressure, the BCM communicates this to the IPC on the vehicle communication network.

Tire Pressure Monitoring System (TPMS) Pressure Sensor

The BCM uses tire pressure sensors to monitor the tire pressure in the 4 road tires. The sensors transmit radio frequency signals to the BCM approximately once every 60 seconds when the vehicle speed exceeds 32 km/h (20 mph).

The tire pressure sensors are battery operated and mounted to the valve stem.

Instrument Panel Cluster (IPC) and Message Center

The IPC illuminates the TPMS warning indicator when it receives a TPMS warning indicator on request from the BCM and displays the appropriate message(s) in the message center.

For additional information on the IPC and message center, refer to Section 413-01.